

Your Phone, Your Network, Your Apps: Everything Would Change by 2020



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Even as buyers wonder whether to buy a 3G phone or a 4G one, even as developers discuss which platform to develop their apps for, even as wearable makers and media houses harness the boons of 4G, elsewhere in the world, researchers are moving ahead.

Major telcos and device makers are betting on the arrival of 5G by 2018, or at the most by 2020. This would mean more than just buying another instrument and changing your data plan. Trend watchers and industry experts believe that this will change everything in the telecom value chain. The network infrastructure would change; bandwidth would be unimaginably high. Millions of users would be able to send across large data packets, say, a movie would transfer before you can say zap! This would mean many changes in mobile devices—not just in terms of wireless technology, but everything from battery and storage to camera and display, because with better connectivity people would want to do much more with their phones. As people do more with their phones, more data would get generated; so data centres would be impacted too. Homes and cities would get smarter. More signals would fly across the air, and perhaps more birds would die—unless technologists come up with a solution for that too.

Here, in this feature, we take a look at some ways in which global telecommunications is bound to change in the next couple of years. Would you be able to recognise the phone in your pocket? Would you buy CDs anymore? Would you pay a smaller or larger mobile bill? Read on to find out.

A network that understands the user's needs

The 5G network will provide speeds that you cannot even imagine today. Prof. Rahim Tafazolli, director of 5G Innovation Centre (5GIC) at University of Surrey, said in a recent BBC News story that it would be possible to run wireless data connections at 800Gbps with 5G, which would, according to him, be a dramatic overhaul and harmonisation of the radio spectrum.

5G will be like the Internet. 5GIC likens the 5G network to the Internet in its ability to adapt, evolve and grow. It will not be a frigid network that chokes and splutters under pressure from smart homes, cars and cities! Despite a constant increase in the number of connected devices, it will attempt to enable real-time responses for all.

Logically unlimited data rates. While such limitless bandwidth is not physically possible, it can be done logically; that is the innovation in 5G. The upcom-



5G Innovation Centre at the University of Surrey (Courtesy: University of Surrey)